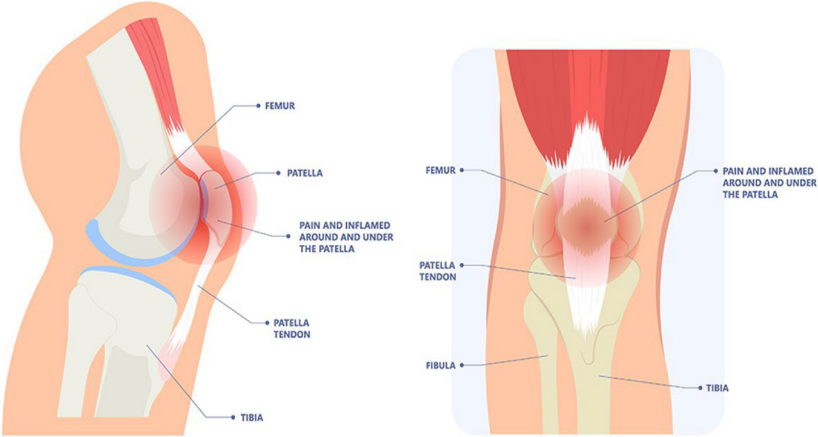
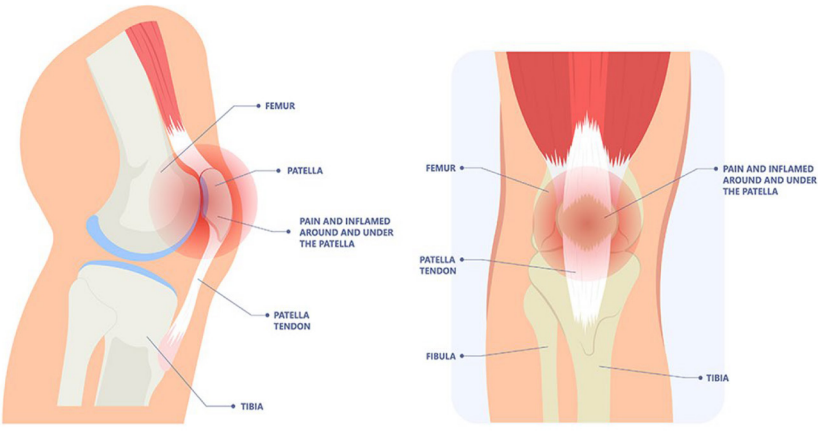


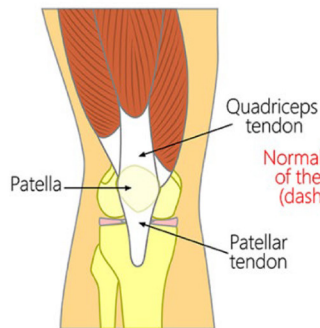
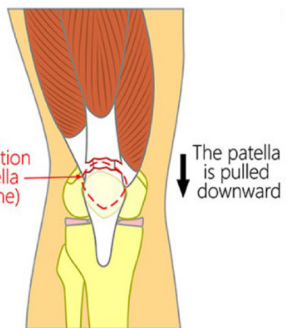
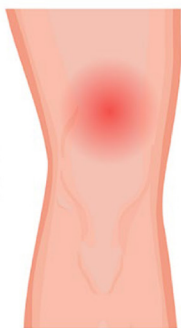
Common Knee Injuries in Bodybuilding



Injury Type	General Information	Symptoms	Relevant Risk Factors
Patellofemoral Pain Syndrome  <i>Adapted from Pepermporn/Shutterstock.com</i>	Proposed to be a result of biomechanical factors and tissue imbalances causing improper tracking of the knee during movement and excessive stress of the patellofemoral joint (where the kneecap and femur bones join) and its chondral (cartilage) surfaces.	Pain in the patellar (kneecap) region that is worse with physical activity especially stair negotiation, jumping, running, and squatting and eccentric loading.	Sudden spikes in training volume; muscle imbalances that may exacerbate knee valgus (knock knees) and foot pronation; muscle weakness of the vastus medialis obliquus; tightness of the iliotibial band, hamstrings, soleus, and gastrocnemius.
Patellar Tendinopathy  <i>Adapted from Pepermporn/Shutterstock.com</i>	Load-dependent pain in either the tendon linking the quadriceps musculature to the patella (quadriceps tendon) or pain in the tendon between the inferior pole of the patella and the tibial tuberosity (patellar tendon).	Pain in the knee (particularly in the inferior patellar tendon) that increases with loading (% 1RM).	No strong risk factors for development of patellar tendinopathy despite some literature showing there may be merit investigating ankle dorsiflexion range of motion, posterior thigh, and quadriceps flexibility. Surprisingly, sudden spikes in training volume are not strongly linked to symptoms.

Common Knee Injuries in Bodybuilding



Injury Type	General Information	Symptoms	Relevant Risk Factors
Quadriceps Tendon Rupture QUADRICEPS TENDON IS INTACT  QUADRICEPS TENDON RUPTURE  PAIN AREA 	A complete tearing (rupture) of the quadriceps muscle tendon from the insertion on the patella or from the patellar tendon's insertion on the tibial tuberosity.	Presence of significant pain and swelling and inability to extend the knee secondary to a fall or traumatic injury.	Chronic microtrauma (e.g., presence of existing tendinopathy); corticosteroid injections; previous ACL repair; use of anabolic steroids.

Adapted from Aksanaku/Shutterstock.com

All data from (Halabchi et al., 2013; Hryvniak et al., 2014; Kanayama et al., 2015; Nori, 2018; Sprague et al., 2019)